Galaxies Quench When Gas Supply Loses the Race

A step-function response from gas-mass conservation unifies black-hole and halo quenching across 350,000 galaxies.

arXiv:2610.20505 — Step-like quenching from gas supply versus depletion timescales

Authors: Zhicheng He, Enci Wang, Luis C. Ho, Yingjie Peng, Huiyuan Wang, Yong Shi , Tao Wang, Yongquan Xue et al.

arXiv:2610.20505 doi:10.48550/arXiv.2610.20505

Background

Galaxy quenching—the shutdown of star formation—arises from many processes, yet no simple unified framework connects them. Gas supply and gas depletion timescales set the outcome.

> **No simple, unified framework links diverse quenching mechanisms to a single observable response.**

What this paper is selling

Gas-mass conservation yields a step-function quenching response, letting the gas-supply timescale be inferred directly from the observed quenched fraction. Applying it to ~350,000 galaxies reveals a characteristic 1 Gyr supply timescale.

Key results to remember

- Supply timescale is ~1 Gyr, inferred directly from the quenched fraction.
- Black holes drive inside-out quenching; dark halos drive outside-in quenching.
- Intermediate surface density marks a critical state for quenching and rejuvenation.

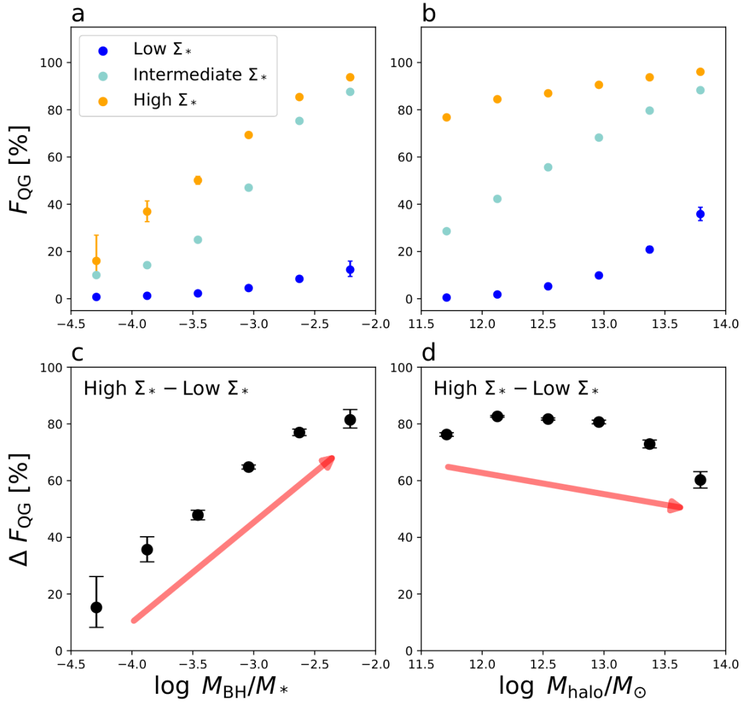


Figure 3 | Quenching responds most sharply to black-hole and halo regulation at intermediate stellar surface density, the critical regime.

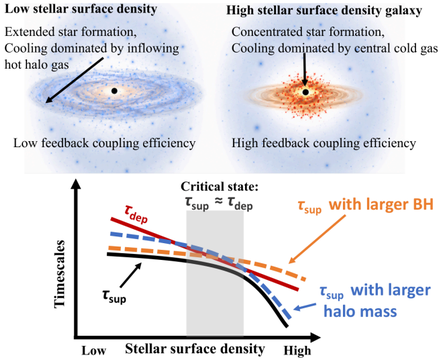


Figure 4 | A physical picture: extended gas supply at low density, centrally concentrated supply at high density, making timescales comparable at intermediate density.